

Now Hack It

To change your game, you'll need to edit the files that FPS Creator packs into it—you'll find these under Program Files\The Game Creators\FPS Creator Demo\Files. The first (and easiest) thing you'll want to do with your game is to change the way it looks and sounds. Little of the rest is easy, and some parts profoundly bizarre. Depending on how deep you want to go, you'll need basic knowledge of image editing and 3D concepts, and some programming knowledge wouldn't hurt, either.

Hear Me

Once you've navigated to the Files folder, you will find the game sounds in the audiobank folder. The organisation is quite simple and self-explanatory, and all you have to do is replace the existing sounds with the ones you really want, or just edit them in an audio-editing program. Our recommendation: first attack the player's extremely annoying jump grunt—audiobank\voices\player\leap.wav. FPSC also lets you customise the way your footsteps and gunshots sound when walking on / shooting at different materials. You'll find them in the audiobank\materials folder. You will find sounds suffixed with either S, I, D, or nothing. The ones without the suffix are files that will be played when you walk on that material.

The "S" sounds are played when something scrapes against the material, "I" for impact sounds like falls, and "D" when the object is destroyed. You'll find these details in materialdefault.txt right there, along with another parameter called matdecals—this is the decal (an animated visual) that shows up every time the sound is played. For example, if you shoot into the ground, you'll hear the impact sound, and the dust decal plays, showing you dust rising from the gunshot.

See Me

We've just talked about decals, so let's start with those. Strictly speaking, a decal is a picture or design applied to a surface. Here, the surface in question is a 3D model—mostly a particle system. FPSC uses the DDS (DirectDrawSurface) file format for all its textures and decals—it occupies very little space in your GPU's memory, because it remains compressed there as well, unlike other formats that uncompress into memory. To view DDS files, all you need to do is download a plugin for IrfanView. To edit them, you will need to download NVIDIA's DDS plugin for Photoshop from http://developer.nvidia.com/object/photoshop_dds_plugins.html. If you don't use Photoshop, you can get DDS Converter from http://eliteforce2.filefront.com/file/DDS_Converter;29412, which converts from most popular formats to DDS. You will find FPS Creator's decals under \gamecore\decals. Each decal is in a folder of its own, with a file called decal-spec.txt, which contains this:

```
;decal spec
;DECAL
across      = 4
down        = 4
offsety     = 50
```

The first two lines are just comments; the remaining is what matters. If you open up the decal, you'll see it divided into sixteen squares—one for each frame of the animation. "across" and "down" are the number of squares that you've drawn in those directions, and "offsety" is how much the decal should be raised above the ground. If you set it to zero for the fire decal, for example, you would end up seeing the top of the fire just brushing the floor surface. Getting it right is a matter of trial and error. The black part of the decal is actually the Alpha channel, which determines what parts of the decal are transparent. Naturally, you'll need an image editor that supports alpha channels—this applies for all the DDS files in your game.

If you've made a new decal and want to incorporate it into the game, you also need to enter it into the entity bank. Go into the \entitybank\generic\decals folder, make a copy of one of the .fpe files there, and give it the name you want to give your decal. Open it up and you'll see this:

```
;header
desc      = {put a one-word
description for your decal here}

;ai
aiinit     = repeatsound.fpi
{if you don't want a sound to play
around the decal, use appear.fpi}
aimain     = decal.fpi
aidestroy  = disappear1.fpi

;orientation
model      = particles.x
textured    = particles.d.tga
transparency = 0
defaultheight = 0.5
```



What happens if you forget the Alpha channel



Ah, the decal is much better this way

